Workflow Builder API Documentation

This document explains how to use the Workflow Builder API. It is designed to be simple and easy to understand.

1. Main Components

The API has these main parts: * **Authentication**: Checks who you are. * **Workflows**: Create and list your workflows. * **Nodes & Edges**: Build the steps and connections in a workflow. * **Actions & Categories**: The building blocks you can use in a workflow.

2. API Response Explained

Here is a simple explanation of what the API sends back (Example: A single workflow).

```
{
  "id": "123-abc",
  "name": "My First Workflow",
  "createdAt": "2023-01-01T12:00:00Z",
  "nodes": [],
  "edges": []
}
```

- `id`: The unique ID of the workflow.
- `name`: What you named the workflow.
- `createdAt`: When it was created.
- `nodes`: A list of steps involved.
- `edges`: A list of connections between steps.

3. How to Use an Endpoint

Endpoint: `GET /workflows` **Purpose**: Gets a list of all your workflows.

To use it, you send a request to the server address with `/workflows` at the end. You must include your "Key" (Token) so the server knows it is you.

4. Simple Code Example

Here is how you can use the API with JavaScript:

```
const response = await fetch('https://api.lekana.ai/workflows', {
  method: 'GET',
  headers: {
    'Authorization': 'Bearer YOUR_TOKEN_HERE',
    'Content-Type': 'application/json'
  }
```

```
});

const data = await response.json();
console.log(data);
```

5. Rate Limits and Authentication

- **Authentication**: You need a "Bearer Token". This is like a digital ID card. You put it in the header of your request.
- **Rate Limits**: The system limits how many times you can ask for data in a short time to keep it fast for everyone. (Specific limits are managed by the server settings).

6. Jargon Simplified

- **Endpoint**: A specific web address for a function (like a "page" for a computer).
- **Payload**: The data you send to the server.
- **JSON**: A text format used to send data that is easy for computers to read.

7. Key Points Summary

- This API lets you build and save workflows.
- You must be logged in.
- It uses JSON for all data.

8. Assumptions

The documentation assumes: * You have an account and a valid token. * You know the basic URL of the API server. * You are sending valid data formats.

9. Missing Information

- Specific limits on how many nodes a workflow can have.
- Exact rate limit numbers (e.g., 60 requests per minute).

10. Key Takeaways

- Always use headers for Auth.
- Check the `401` error if things don't work (it usually means auth issues).
- Workflows are the main object you will interact with.

11. Next Steps

1. Get your bearer token from the login page.
2. Try the `GET /workflows` endpoint to see if it works.

3. Try creating a workflow with `POST /workflows`.

12. Error Codes

- **400 Bad Request**: You sent missing or wrong data (e.g., no name).
- **401 Unauthorized**: You didn't provide a valid token.
- **403 Forbidden**: You don't have permission for this.
- **404 Not Found**: The thing you asked for doesn't exist.
- **500 Internal Server Error**: Something went wrong on the server side.

13. Data Types

- **String**: Text (e.g., "My Workflow").
- **Boolean**: True or False.
- **JSON**: A nested object of data (used for configurations).
- **UUID**: A long random string used as a unique ID.

14. Best Practices

- **Cache your data**: Don't ask for the same workflow 10 times in a second.
- **Handle Errors**: Always write code to check if the request failed.
- **Secure your Token**: Never share your token with others.

15. Common Mistakes

- Forgetting to add the `Authorization` header.
- Sending the wrong ID format (it must be a valid ID).
- Sending JSON data without setting `Content-Type: application/json`.

16. Edge Cases to Handle

- What if the workflow list is empty?
- What if you try to add a node to a workflow that was just deleted?
- What if the server is down?

17. Suggested Improvements

- Add **Pagination**: If a user has 1000 workflows, send them in pages of 20.
- Add **Filtering**: Allow searching workflows by name.

18. Final API Reference

Method	Endpoint	Description
GET	`/workflows`	List all workflows.

Method	Endpoint	Description
POST	`/workflows`	Create a new workflow. Body: `{ "name": "..." }`
GET	`/workflows/:id`	Get details of one workflow.
POST	`/workflows/:id/nodes`	Add a node.
POST	`/workflows/:id/edges`	Add a connection.
GET	`/actions`	List all available actions.
GET	`/categories`	List all action categories.

19. Sample Requests

Here are examples of how to call the main endpoints using cURL. You can import these into Postman.

A. Create a Workflow

```
curl -X POST https://api.lekana.ai/workflows \
  -H "Authorization: Bearer YOUR_TOKEN_HERE" \
  -H "Content-Type: application/json" \
  -d '{
    "name": "Document Processing Pipeline"
  }'
```

B. Add a Node to a Workflow

- **Requires**: `actionId` (from GET /actions), `positionX`, `positionY`.

```
curl -X POST https://api.lekana.ai/workflows/YOUR_WORKFLOW_ID/nodes \
  -H "Authorization: Bearer YOUR_TOKEN_HERE" \
  -H "Content-Type: application/json" \
  -d '{
    "actionId": "uuid-of-action",
    "positionX": 100,
    "positionY": 200,
    "config": {
      "format": "pdf"
    }
  }'
```

C. Connect Two Nodes (Create Edge)

- **Requires**: `sourceNodeId` and `targetNodeId`.

```
curl -X POST https://api.lekana.ai/workflows/YOUR_WORKFLOW_ID/edges \
  -H "Authorization: Bearer YOUR_TOKEN_HERE" \
  -H "Content-Type: application/json" \
  -d '{
    "sourceNodeId": "uuid-node-A",
    "targetNodeId": "uuid-node-B"
  }'
```

D. Get All Workflows

```
curl -X GET https://api.lekana.ai/workflows \
  -H "Authorization: Bearer YOUR_TOKEN_HERE"
```

Note on PDF: To save this as a PDF, most code editors (like VS Code) allow you to "Export to PDF" using a plugin (like Markdown PDF), or you can open this file in a browser (after converting to HTML) and choose **Print > Save as PDF**.